

NARENTHIRAN MATHAVARAJ

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[Portfolio Website](#)

EDUCATION

National Institute of Technology, Tiruchirappalli

B.Tech in Electronics and Communication Engineering - CGPA: 8.53

Jul 2021 - May 2025

Tiruchirappalli, Tamil Nadu

Pushpalatha Vidya Mandir

All India Senior School Certificate Examination - Percentage: 95.4

May 2021

Tirunelveli, Tamil Nadu

PATENT AND PUBLICATION

- Filed a **Provisional Patent [Patent Application Number: 202441043076]** in the **Indian Patent Office** with **Dr. Nirav Patel** for a Constraint Constant Velocity (CCV) Joint that separates **translational and rotational motion** of the end effector of Master Tool Manipulator (MTM) used in **minimally invasive surgeries**.
- Presented a research paper titled “**Spider Arm: Design and fabrication of a cost-effective six-axis robotic arm featuring a novel double-ring load transfer mechanism**” at the 7th International Conference on Mechatronics, Robotics, and Automation (ICMRA), Wuhan, China September 21st, 2024 ([Link](#)) and published on **IEEE Xplore** ([Paper Link](#)) .

TECHNICAL SKILLS

Languages: Python, C, Rospy

Engineering Software: Fusion 360, LT Spice

Frameworks: ROS1, ROS2, Gazebo, OpenCV, Moveit, Rviz, pybullet, TensorFlow, Stable-Baselines3, Jupyter Notebook

Boards: Arduino Uno, Esp32, Esp8266, Raspberry PI 3, Teensy 4.1, eYFI board

INTERNSHIP EXPERIENCE

Robotics Software Internship

Sept 2025 - Present

Versor [Artpark - IISC], Karnataka

- Implemented a **Follow-Me navigation system** for agricultural environments using person detection via **YOLOv11** and **YOLOe** with ByteTrack enabled and collision avoidance with **ROS2 Nav2 Behaviour Trees**, which can handle partial occlusions, enabling autonomous operation in unstructured farm terrain without a pre-existing map.
- Developing a sensor fusion pipeline to integrate **Livox Mid-360 LiDAR** and multi-camera data, performing LiDAR-camera calibration and frame alignment.
- Trained a reinforcement learning agent using **Proximal Policy Optimization (PPO)** to classify non-harmful obstacles and generate an adaptive navigation costmap, improving performance over height-map-based methods in off-road terrain navigation.
- Working Videos ([Link](#), [Link](#)).

Summer Research Internship

Jun 2024 - Aug 2024

(Under Dr. Nirav Patel)

IIT-Madras, Tamil Nadu

- Worked as a **research** intern in the Engineering Design Department at **IIT-Madras** on a medical robot that aims to reduce the cost of **minimally invasive surgeries**.
- Designed, and 3D printed a provisional patent filed **Low-Cost Master Tool Manipulator (MTM)** in **Fusion 360**, which allows the surgeon to control the tooltip of the slave device **Patient Side Manipulator (PSM)** for precise manipulation.
- Worked on real-time communication for **teleoperation** between master (MTM) and slave device (PSM) via **UDP protocol**.
- Wrote the **Forward Kinematics** script in Python for the **4-DOF master controller** (MTM) and wrote ESP32 script to transfer the encoder's value from the master controller to **Rospy** via Wifi (UDP Protocol) and serial communication.
- Wrote an **Inverse Kinematics** script to control the **ST servo's roll, pitch, and yaw** motion in **PSM**.
- Wrote a Python script with the **Moveit** package to test and analyze the MTM and PSM in **Rviz**.
- Block Diagram: ([Link](#)).

Summer Research Internship

Jun 2024 - Aug 2024

(Under Dr. Leena Vachhani)

IIT-Bombay, Maharashtra

- Worked as a **research** intern at **IIT-Bombay** on a **Commercial-Off-The-Shelf (COTS)** based robot that aims to do path planning using a drivable area segmentation model that runs offline on a smartphone.
- Worked in Gazebo to obtain depth estimation using **Monocular Depth Estimation**.
- Created **Depth Maps** and **Polar Occupancy Grid** for the vehicles and drivable areas respectively, and evaluated the performance of different segmentation models, including **Deeplab**, **FFNet**, and **DDRNet**, on **COTS devices**, such as smartphones.
- Worked on developing a mobile application to deploy and run a segmentation model (**TFLite model**) on smartphones.

PROJECTS

Deep Q-Learning Based Reinforcement Learning ([Github](#))

Aug 2025

Mini Project

Remote

- Designed and implemented a **Deep Q-Learning (DQN) model** in Jupyter Notebook to balance the pole in CartPole-v0 from Gymnasium, utilizing a 4-layer neural network with ReLU activations and linear output.
- Applied reinforcement learning techniques including **epsilon-greedy exploration** and **soft target network updates** to improve policy stability and convergence.

Turtlesim Turtle Catcher ([Github](#))

Aug 2025

Mini Project

Remote

- Built and tested a **ROS2** workspace by creating **custom nodes** and verifying inter-node communication using **topics**, **publishers**, and **subscribers**.
- Explored **ROS2 parameters** and services to configure node behavior dynamically, ensuring modularity and reusability of system components.
- Conducted system-level debugging and monitoring with **ROS2 CLI** tools (ros2 topic, ros2 node, ros2 param) to validate functionality and improve reliability of robotic applications.

Spider_Arm

Oct 2023 - Jul 2025

Research Club's Project, Spider, R&D Club, NIT Tiruchirappalli

NIT-Trichy, Tamil Nadu

- Designed a cost-effective **6 Degrees of Freedom (6-DOF)** serial manipulator with a reach of **700 mm**, a precision level of **5 mm**, and the capability to handle a **0.5 kg payload** all within the budget of **\$700**.
- Designed the manipulator using **Fusion 360** and worked on the hardware (**Stepper Motors, Limit Switches, Teensy 4.1**) and software interface.
- Simulated the manipulator in the **Gazebo**, visualized using **Rviz**, and proceeded with the motion planning using **Moveit**.
- Completed phase one of fabrication and tested the **4 DOF Assembly** motion and executed basic pick and place in **Pybullet**.
- Awarded Seed Funding of Rs:50,000** by **GUJCOST** (Gujarat Council of Science and Technology) through **Robofest 4.0** (India's Biggest Robotics Competition).

Hola Bot ([Github](#))

Aug 2022 -Feb 2023

E-Yantra Robotics Competition: (Semi - Finalist)

IIT-Bombay, Maharashtra

- Developed a **3-wheel holonomic robot** (Mobile Robot) with **3 Degrees of Freedom (DOF)** to draw complex art.
- Designed the robot using **Fusion 360** and **OpenCV** to detect the **Aruco Marker** to identify the robot's position. The robot's navigation was done by the **Proportional** and **Differential Controller**.
- Simulated the robot in **Gazebo** and wrote **ROS** node to subscribe the positional data of the robot in the canvas and publish the speed required for the three wheels of the robot.
- Worked with custom eYFi-Mega Board (**Atmega2560+Esp32**) board to connect to **ROS** node via **Wifi** and control the speed of the stepper motors.

ACADEMIC ACHIEVEMENTS

- Secured **Excellent Presenter Award** ([Link](#)) at the 7th International Conference on Mechatronics, Robotics, and Automation (ICMRA), Wuhan, China on **September 21st, 2024**.
- Secured **Rank 23** in **National Mathematics Indian Talent** at the State level conducted by Indian Talent in **2017**.
- Was among the top **10%** of the candidates in the **International Olympiad of Mathematics** conducted by Silver Zone Olympiad in **2015**.

POSITION OF RESPONSIBILITY

Spider Research and Development Club

Aug 2022 - July 2025

Researcher

NIT-Trichy, Tamil Nadu

- Spider R&D** is the official Research and Development Club of **NIT, Trichy**.
- As a member and an active researcher, worked on club **research projects** and participated in **competitions** and **hackathons**.
- Doing Research projects in the field of **mobile robotics** and **arm manipulation**.
- Currently, published a research paper on **6-Degrees Of Freedom (DOF)** robotic arm **SPIDER_ARM**.

Festember Marketing Team

Dec 2022 - Oct 2023

Manager

NIT-Trichy, Tamil Nadu

- Festember** is South India's Largest Cultural Festival conducted by **NIT, Trichy**.
- As part of this team, coordinated with sponsors and partners from different **MNCs** to collect capital for running a successful festival.

EXTRACURRICULAR ACTIVITIES

- Conducted a robotics workshop called **Techids**, for underprivileged school students as part of **Pragyan Social Responsibility Event** in **2022**.
- Secured **Second Place** in **KKCA 2nd All India FIDE Rated Chess Tournament** organized by **Kovai Kids Chess Academy** in **2017**.